
Foundation Models for Temporal Systems: From Forecasting to World Modeling

Boris N. Oreshkin
Amazon

Danielle Maddix Robinson
AWS

Omri Azencot
Ben-Gurion University

Ming Jin
Griffith University

Emadeldeen Eldele
Khalifa University

Mayank Jauhari
Amazon

Chenghao Liu
Datadog

N. Benjamin Erichson
Berkeley Lab

1 Workshop Overview, Motivation, and Relevance

Workshop pitch. This workshop convenes the time-series machine learning community around temporal world modeling: foundation models for forecasting, simulation, scenario analysis, and reliable decision-making in complex temporal systems. We focus on settings where temporal data is multimodal, asynchronous, event-driven, and multi-scale, and where models must reason beyond extrapolation under distribution shift. We solicit contributions across four directions: (i) forecasting and simulation tasks, (ii) temporal data and environments, (iii) temporal models, and (iv) evaluation and reliability, with applications in climate, healthcare, and industrial forecasting.

Background and Motivation. Temporal machine learning is shifting from task-specific forecasting toward foundation models for complex temporal systems. Progress in large-scale temporal pretraining [2, 12], multi-time-scale world models [22], multimodal temporal modeling [16], and generative simulation [18] remains fragmented across communities, despite a shared target: real-world temporal systems that are multimodal, asynchronous, event-driven, multi-scale, and distribution-shifted [7, 10]. Standard benchmarks and the models they shaped [31, 27, 17, 1, 23] drive predictive accuracy, but not the broader requirements of temporal world modeling: robustness, calibration, long-horizon consistency, simulation fidelity, causal reasoning, and interaction with external context.

Workshop Objective. The workshop will develop this framing into a research agenda for coherent modeling of evolving real-world temporal systems, extending the world-models lineage of Ha and Schmidhuber [13] to **temporal, multimodal, and decision-relevant settings**. It brings together time-series machine learning, multimodal learning, generative temporal modeling, foundation models, and adaptive systems around shared benchmarks, datasets, modeling paradigms, and **evaluation protocols** for robustness, calibration, long-horizon consistency, and reproducibility under distribution shift. Once a model is treated not only as a forecaster but as a simulator of an evolving system, **agentic mechanisms** such as model selection, ensembling, retrieval, and context conditioning become part of *operating* the world model rather than separate downstream pipelines. We translate this objective into four connected axes, expanded below: tasks, data and environments, models, and evaluation protocols for reliable temporal world modeling.

Forecasting and Simulation Tasks. This axis defines the task surface for temporal world models beyond fixed-horizon prediction: long-horizon and multi-resolution forecasting; multimodal contextual forecasting conditioned on text, video, events, covariates, or actions; probabilistic forecasting with calibrated uncertainty; prediction under sparse observations, regime changes, and distribution shift; and scalable trajectory simulation under heterogeneous conditions. We also solicit adaptive, retrieval-augmented, and agentic forecasting systems, including in-context fine-tuning [8], retrieval-augmented forecasting [25], diffusion-transformer forecasting [6], model selection, ensembling, tool use, planning, and inference-time orchestration as mechanisms for *operating* temporal world models at deployment time.

Temporal Data and Environments. This axis focuses on the data substrate for temporal world modeling: moving from isolated benchmarks to richer *temporal environments* that combine time series with text, video, graphs, sensors, trajectories, actions, and events. Representative directions include large-scale temporal pretraining corpora [24, 26], video-as-environment corpora [3, 5, 30], long-horizon physical evaluation suites [21], synthetic temporal data generation [2, 9], simulation environments, benchmark realism, data quality, and scalable evaluation. The workshop treats datasets, simulators, and temporal environments, not only models, as primary research artifacts for building and evaluating temporal world models.

Temporal Models. This axis focuses on model classes for representing evolving temporal systems: time-aware models for irregular sampling and variable step sizes [19]; event-based models for marked temporal point processes and transaction streams [4, 29]; hierarchical multi-timescale and state-space architectures for long-horizon dynamics [22]; generative temporal models for trajectory prediction, simulation, and video-as-world-model settings; and time-series foundation models (TSFMs) [2, 12, 26, 24]. Additional topics include scalable pretraining, multimodal temporal fusion, memory and persistent-state representations, embodied vision-language-action systems, and adaptive forecasters across domains, modalities, and horizons.

Evaluation and Reliability. This axis focuses on evaluation protocols for temporal world models beyond point accuracy: benchmark realism, data quality, and leakage-aware evaluation; robustness under distribution shift and test-time adaptation [14]; calibration and uncertainty quantification via conformal methods [28, 11]; long-horizon and simulation consistency, including arbitrage-based metrics [20]; neural scaling laws [24]; and reproducibility standards for temporal datasets, pretraining corpora, and benchmark contamination audits for time-series foundation models [15].

2 Invited Speakers and Audience

The workshop targets researchers and practitioners across time-series machine learning, foundation models, multimodal and generative temporal modeling, scientific machine learning, and world models, alongside industry teams building production forecasting and simulation systems for healthcare, climate, and operations. The confirmed speaker slate, detailed below, spans these communities across academia and industry.

Rose Yu (confirmed) is an Associate Professor of Computer Science at UC San Diego and the Halicioğlu Data Science Institute, and an Amazon Scholar. Her research develops physics-guided deep learning for spatiotemporal data, including generative models for spatiotemporal sequences and physics-informed neural networks. Her work on multimodal LLM agents and irregular temporal structures engages the workshop’s themes of inductive biases and missing-data handling.

Michael W. Mahoney (confirmed) is a Professor of Statistics at UC Berkeley, Director of the Big Data Group at ICSI, and Amazon Scholar. He pioneered randomized numerical linear algebra and collaborated on the Chronos and Chronos-2 time-series foundation models. His work on scaling and transfer behaviour of foundation models for scientific machine learning engages the workshop’s themes of scaling laws and evaluation rigor.

Abdul Fatir Ansari (confirmed) is a Senior Applied Scientist at AWS working on large-scale TSFMs. He leads development of the Chronos family, with prior work on diffusion-based probabilistic forecasting. His work on tokenization-based pretraining and zero-shot transfer engages the workshop’s themes of foundation-model design and operationalization.

Aditi Krishnapriyan (confirmed) is the Stangeland Assistant Professor of Chemical and Biomolecular Engineering and EECS at UC Berkeley, with appointments in BAIR and at LBNL. Her research develops physics-inspired machine learning integrating physical inductive biases, geometric structure, and differentiable physics, with applications to fluid dynamics and neural PDE solvers. Her work engages the workshop’s themes of physics-informed temporal modeling and reliability on continuous-time dynamics.

Marinka Zitnik (confirmed) is an Associate Professor of Biomedical Informatics at Harvard Medical School and founder of Therapeutics Data Commons. Her group co-developed UniTS, a multi-task time-series model handling classification, forecasting, imputation, and anomaly detection. Her work on multi-task temporal foundation models and biomedical time-series heterogeneity engages the workshop’s themes of multimodal temporal learning and reliability in safety-critical domains.

Morning		Afternoon	
09:00–09:10	Opening remarks	13:30–14:10	Invited talks VI–VII
09:10–10:10	Invited talks I–III	14:10–14:55	Coffee break and Posters II
10:10–10:55	Coffee break and Posters I	14:55–15:35	Invited talks VIII–IX
10:55–11:35	Invited talks IV–V	15:35–16:20	Spotlights II: 5 talks + Q&A
11:35–12:20	Spotlights I: 5 talks + Q&A	16:20–17:00	Moderated panel
12:20–13:30	Lunch	17:00–17:25	Open community discussion
		17:25–17:30	Closing remarks

Table 1: Tentative one-day schedule; timing will be adjusted to NeurIPS 2026 logistics. Invited talks are 20 minutes each; spotlight talks are 7 minutes plus Q&A.

Tim Januschowski (confirmed) is a long-standing contributor to time-series machine learning and open-source forecasting infrastructure at Databricks. He co-developed GluonTS and is lead author of foundational works on production forecasting. His perspective on bridging research and deployment engages the workshop’s themes of evaluation realism, reliability, and operationalization.

Daniel F. Schmidt (confirmed) is an Associate Professor of Data Science at Monash University. He co-authored the MONSTER benchmark for scalable time-series classification and SETAR-Tree, a global tree-based forecasting method. His work on principled model selection and information-theoretic complexity engages the workshop’s themes of benchmark realism and evaluation rigor.

Mingsheng Long (confirmed) is a tenured Associate Professor in the School of Software at Tsinghua University. He has led development of widely used time-series architectures including Autoformer, TimesNet, iTransformer, and the Anomaly Transformer, and maintains TSlib as a community-standard library. His work on transferable temporal representations engages the workshop’s themes of foundation-model design and reproducible evaluation.

Flora Salim (confirmed) is a Professor of Computer Science at UNSW Sydney, holds the Cisco Chair of Digital Transport, and is Deputy Director of the UNSW AI Institute. Her research focuses on data-efficient learning with multimodal sensor data, spatio-temporal modeling, and continual learning, applied to transport and urban systems. Her work engages the workshop’s themes of multimodal forecasting and distribution-shift-aware evaluation.

3 Community Building and Workshop Format

The workshop will connect fragmented academic and industry communities working on temporal forecasting, simulation, and world modeling, with applications in climate, healthcare, and industrial systems. The program combines invited talks, contributed papers, posters, spotlights, a moderated panel, and an open discussion on benchmark realism, distribution shift, simulation fidelity, and foundation models for temporal systems. Contributed papers will be managed through OpenReview under standard conflict-of-interest and review policies, and reviewed by the organizing team and program committee. Poster and spotlight selections will prioritize technical quality, topical diversity, and interaction across time-series machine learning, multimodal learning, foundation models, and generative temporal modeling.

The tentative one-day program is summarized in Table 1. We anticipate approximately 200-300 in-person attendees, based on recent NeurIPS and ICML time-series workshops. The preferred venue order is Sydney, Paris, Atlanta, with no requirements beyond standard audiovisual support. This is a new workshop, not a continuation. It complements adjacent venues (*Time Series in the Age of Large Models*, *BERT2S*, *MiLeTS*, *AI4TS*) while distinguishing itself by unifying forecasting and simulation, multimodal temporal environments, agentic inference-time operation, and event-based and irregular modeling: what *temporal world modeling* adds beyond forecasting.

Tagline: Foundation models for temporal world modeling: forecasting, simulation, and reliability.

4 Diversity, Inclusion, and Accessibility

The workshop is committed to a diverse and inclusive research community, and the slate reflects this concretely. Women make up 4 of the 9 invited speakers (44%): Rose Yu, Aditi Krishnapriyan, Marinka Zitnik, and Flora Salim. The team draws on globally distributed institutions across Asia, the Middle East, Australia, Europe, and North America, over 80% of non-North-American origin. It bridges sectors, with 6 industry and 11 academic/public-research affiliations (35% industry), and balances 3 senior, 4 mid-career, and 2 junior researchers. The workshop will prioritise junior and early-career contributions and follow standard NeurIPS accessibility and inclusion guidelines.

Organizer Information

None of the organizers are organizing other NeurIPS 2026 workshops concurrently, except Danielle Maddix Robinson, who is a co-organizer on “From Scientific Foundation Models to Agentic AI Scientists: Reliable and Reusable AI for Computational Discovery.”

Boris N. Oreshkin, Amazon, oreshkin@amazon.com. Principal Scientist at Amazon specializing in time series forecasting, probabilistic modeling, and foundation models for temporal data. He is the lead author of N-BEATS and a pioneer of zero-shot time series forecasting. His research focuses on large-scale forecasting systems within Amazon Supply Chain Optimization Technologies (SCOT). Prior to Amazon, he held research positions at Unity Labs, Apple, and Element AI. He received his PhD in Electrical Engineering from McGill University.

Danielle Maddix Robinson, AWS, dmmaddix@amazon.com. Senior Applied Scientist at AWS specializing in time series forecasting, scientific machine learning, and physics-constrained foundation models. She leads the DeepEarth team focused on physics-informed machine learning for scientific computing and received her PhD in Computational and Mathematical Engineering from Stanford University. She previously co-organized the ICLR 2024 Workshop on AI for Differential Equations in Science and received the Best Paper Award at the NeurIPS 2020 ML and Public Health Workshop. Danielle is also a co-organizer on “From Scientific Foundation Models to Agentic AI Scientists” (NeurIPS 2026).

Omri Azencot, Ben-Gurion University of the Negev, azencot@bgu.ac.il. Assistant Professor of Computer Science at Ben-Gurion University. His research focuses on generative and representation learning for sequential data, including diffusion models, Koopman dynamics, and irregularly sampled time series. He has published at NeurIPS, ICML, ICLR, and SIGGRAPH, including recent work on generative temporal modeling and time series learning under data scarcity.

Ming Jin, Griffith University, ming.jin@griffith.edu.au Assistant Professor at Griffith University specializing in time series forecasting, foundation models, and spatio-temporal data mining. He is the lead author of Time-LLM, Time-MoE, and TimeMixer++. He has authored over 70 publications with more than 7,000 citations and has extensive experience organizing workshops and tutorials at KDD, AAAI, and WWW.

Emadeldeen Eldele, Khalifa University, emadeldeenahmed.eldele@ku.ac.ae Assistant Professor at Khalifa University specializing in time series representation learning, self-supervised learning, and temporal modeling under distribution shifts. He previously held a research scientist position at A*STAR, Singapore, and received his PhD from Nanyang Technological University. His work has appeared at ICML, NeurIPS, AAAI, and TPAMI, and he received the IEEE Engineering in Medicine and Biology Prize Paper Award in 2023.

Mayank Jauhari, Amazon, jauharim@amazon.com. Senior Applied Scientist at Amazon working on large-scale machine learning systems for finance automation and industrial AI applications. His research interests include time series analysis, anomaly detection, representation learning, and foundation models for temporal data. His applied experience with scalable forecasting and temporal machine learning systems contributes an important industry perspective to the workshop.

Chenghao Liu, Datadog, twinsken@gmail.com. Research Scientist at Datadog specializing in large-scale forecasting systems, anomaly detection, and machine learning for cloud infrastructure and observability platforms. His research focuses on robust temporal modeling under noisy and heterogeneous operational environments, with applications to monitoring and reliability systems.

N. Benjamin Erichson, Berkeley Lab, erichson@lbl.gov. Staff Scientist at the Lawrence Berkeley National Laboratory working on models and methods for processing sequential data, with a particular interest in continuous-time formulations, state-space models, linear attention mechanisms, and neural architectures inspired by dynamical systems. His work has appeared at ICML, NeurIPS, ICLR and AISTATS. He previously co-organized the NeurIPS 2024 and ICLR 2026 Workshop on Foundation Models for Science.

Program Committee and Reviewers. The program committee comprises Rose Yu (UC San Diego), Michael Mahoney (UC Berkeley), Abdul Fatir Ansari (AWS), Tim Januschowski (Databricks), Daniel F. Schmidt (Monash), and Flora Salim (UNSW Sydney). We have confirmed 40+ reviewers spanning academia and industry, including Mononito Goswami (Amazon), Gerald Woo (Datadog), Taha Aksu (Salesforce), Oleksandr Shchur (AWS), Andreas Auer (ELLIS), Arjun Ashok (Mila), Pablo Montero-

Manso (University of Sydney), Haoxin Liu (Georgia Tech), Antonios Valkanas (McGill), Parth Nobel (Stanford), Kareem Hegazy (UC Berkeley), Pu Ren (LBNL), Boran Han (AWS), Xiyuan Zhang (AWS), Shuai Zhang (AWS), Hilaf Hasson (Cohesity), Youngsuk Park (AWS), Syama Rangapuram (AWS), Pedro Mercado (AWS), Michael Bohlke-Schneider (AWS), Jan Gasthaus (Meta), Liran Nochumsohn (Ben-Gurion), Raz Marshanski (Ben-Gurion), and Israel Zexer (Ben-Gurion), among others. Anticipating approximately 80 submissions, this pool supports three reviews per paper with each reviewer assigned at most four to five papers; additional reviewers will be recruited as needed.

References

- [1] T. Aksu, G. Woo, J. Liu, X. Liu, C. Liu, S. Savarese, C. Xiong, and D. Sahoo. Gift-eval: A benchmark for general time series forecasting model evaluation. *arXiv preprint arXiv:2410.10393*, 2024.
- [2] A. F. Ansari, L. Stella, A. C. Türkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Ranganuram, S. Pineda-Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024. arXiv:2403.07815.
- [3] M. Assran, A. Bardes, D. Fan, Q. Garrido, R. Howes, M. Komeili, M. Muckley, A. Rizvi, C. Roberts, K. Sinha, et al. V-JEPA 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*, 2025.
- [4] D. Berghaus, P. Seifner, K. Cvejowski, C. Ojeda, and R. J. Sánchez. In-context learning of temporal point processes with foundation inference models. In *International Conference on Learning Representations (ICLR)*, 2026.
- [5] J. Bruce, M. D. Dennis, A. Edwards, J. Parker-Holder, Y. Shi, E. Hughes, M. Lai, A. Mavalankar, R. Steigerwald, C. Apps, et al. Genie: Generative interactive environments. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.15391.
- [6] D. Cao, W. Ye, Y. Zhang, and Y. Liu. TimeDiT: General-purpose diffusion transformers for time series foundation model. *arXiv preprint arXiv:2409.02322*, 2024.
- [7] C. Chang, J. Hwang, Y. Shi, H. Wang, W.-C. Peng, T.-F. Chen, and W. Wang. Time-IMM: A dataset and benchmark for irregular multimodal multivariate time series. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2025.
- [8] A. Das, M. Faw, R. Sen, and Y. Zhou. In-context fine-tuning for time-series foundation models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. arXiv:2410.24087.
- [9] S. Dooley, G. S. Khurana, C. Mohapatra, S. V. Naidu, and C. White. ForecastPFN: Synthetically-trained zero-shot forecasting. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2311.01933.
- [10] J.-C. Gagnon-Audet, K. Ahuja, M.-J. Darvishi-Bayazi, P. Mousavi, G. Dumas, and I. Rish. WOODS: Benchmarks for out-of-distribution generalization in time series. *arXiv preprint arXiv:2203.09978*, 2022.
- [11] I. Gibbs and E. J. Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. arXiv:2106.00170.
- [12] M. Goswami, K. Szafer, A. Choudhry, Y. Cai, S. Li, and A. Dubrawski. MOMENT: A family of open time-series foundation models. In *Forty-first International Conference on Machine Learning, ICML*, Proceedings of Machine Learning Research, pages 16115–16152, 2024.
- [13] D. Ha and J. Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2018.
- [14] H. Kim, S. Kim, J. Mok, and S. Yoon. Battling non-stationarity in time series forecasting via test-time adaptation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 17868–17876, 2025.
- [15] H. Li, S. Xie, L. Shen, Z. Li, M. Chen, X. Zhang, H. Fu, J. Sun, X. Ren, and C. Liu. TSFMAudit: Data contamination auditing in forecasting time series foundation models. *arXiv preprint arXiv:2605.26161*, 2026.
- [16] H. Liu, S. Xu, Z. Zhao, L. Kong, H. Kamarthi, A. B. Sasanur, M. Sharma, J. Cui, Q. Wen, C. Zhang, and B. A. Prakash. Time-MMD: Multi-domain multimodal dataset for time series analysis. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2024.

- [17] S. Makridakis, E. Spiliotis, and V. Assimakopoulos. The M4 competition: 100,000 time series and 61 forecasting methods. *International Journal of Forecasting*, 36(1):54–74, 2020.
- [18] S. S. Narasimhan, S. Agarwal, O. Akcin, S. Sanghavi, and S. Chinchali. Time weaver: A conditional time series generation model. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.
- [19] A. N. Nhu, S. Son, and M. Lin. Time-aware world model for adaptive prediction and control. In *Forty-second International Conference on Machine Learning, ICML*, Proceedings of Machine Learning Research, 2025.
- [20] D. Paleka, A. Pallavi Sudhir, A. Alvarez, V. Bhat, A. Shen, E. Wang, and F. Tramèr. Consistency checks for language model forecasters. arXiv preprint arXiv:2412.18544, 2024.
- [21] S. Rasp, S. Hoyer, A. Merose, I. Langmore, P. Battaglia, T. Russell, A. Sanchez-Gonzalez, V. Yang, R. Carver, S. Agrawal, et al. WeatherBench 2: A benchmark for the next generation of data-driven global weather models. *Journal of Advances in Modeling Earth Systems*, 16(6), 2024. arXiv:2308.15560.
- [22] V. Shaj Kumar, S. Gholam Zadeh, O. Demir, L. Douat, and G. Neumann. Multi time scale world models. *Advances in Neural Information Processing Systems*, 36:26764–26775, 2023.
- [23] O. Shchur, A. F. Ansari, C. Turkmen, L. Stella, N. Erickson, P. Guéreron, M. Bohlke-Schneider, and Y. Wang. fev-bench: A realistic benchmark for time series forecasting. arXiv preprint arXiv:2509.26468, 2025.
- [24] X. Shi, S. Wang, Y. Nie, D. Li, Z. Ye, Q. Wen, and M. Jin. Time-MoE: Billion-scale time series foundation models with mixture of experts. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. arXiv:2409.16040.
- [25] K. Tire, E. O. Taga, M. E. Ildiz, and S. Oymak. Retrieval-augmented time series forecasting. arXiv preprint arXiv:2411.08249, 2024.
- [26] G. Woo, C. Liu, A. Kumar, C. Xiong, S. Savarese, and D. Sahoo. Unified training of universal time series forecasting transformers. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.02592.
- [27] H. Wu, J. Xu, J. Wang, and M. Long. Autoformer: Decomposition transformers with autocorrelation for long-term series forecasting. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. arXiv:2106.13008.
- [28] C. Xu and Y. Xie. Conformal prediction interval for dynamic time-series. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021. EnbPI; arXiv:2010.09107.
- [29] S. Xue, X. Shi, Z. Chu, Y. Wang, H. Hao, F. Zhou, C. Jiang, C. Pan, J. Y. Zhang, Q. Wen, J. Zhou, and H. Mei. EasyTPP: Towards open benchmarking temporal point processes. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.
- [30] S. Yang, Y. Du, K. Ghasemipour, J. Tompson, L. Kaelbling, D. Schuurmans, and P. Abbeel. Learning interactive real-world simulators. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.06114; UniSim.
- [31] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 11106–11115, 2021. arXiv:2012.07436.